Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	30 September 2026
Team ID	SWTID-2026-2365
Project Name	AI FitTrack API – Fitness Tracking API with Google Gemini AI
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Project Setup	USN-1	As a developer, I can set up the Node.js / Express server, MongoDB connection and MVC folder structure so the team has a working project base.	5	High	Srinivasan K
Sprint-1		USN-2	As a developer, I can add centralised error handling, request logging and environment configuration so failures are handled consistently.	3	Medium	Srinivasan K
Sprint-1	Registration	USN-3	As a user, I can register by entering my name, email and password so that I have my own account.	3	High	Dillibabu J
Sprint-1		USN-4	As a user, I receive a secure JWT token after registering so I can use the application straight	3	High	Dillibabu J

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
			away.			
Sprint-1	Login	USN-5	As a user, I can log in by entering my email and password to access my data.	3	High	Akash S
Sprint-1		USN-6	As a user, I can view my profile details after logging in.	2	Medium	Akash S
Sprint-2	Workout Management	USN-7	As a user, I can add a workout with name, category, duration, calories burned and date.	3	High	Akash S
Sprint-2		USN-8	As a user, I can view all my logged workouts sorted by date (latest first).	2	High	Akash S
Sprint-2		USN-9	As a user, I can view a single workout by its ID.	2	Medium	Mohanasundaram G
Sprint-2		USN-10	As a user, I can update the details of a workout I logged.	3	Medium	Mohanasundaram G
Sprint-2		USN-11	As a user, I can delete a workout I no longer need.	2	Medium	Mohanasundaram G
Sprint-2		USN-12	As a user, I can see only my own workouts so my data stays private.	3	High	Srinivasan K
Sprint-3	Workout Search	USN-13	As a user, I can search my workouts by name or category to find them quickly.	3	Medium	Jagadeesh S
Sprint-3		USN-14	As a user, I can search my workouts by a specific date.	3	Medium	Jagadeesh S
Sprint-3	AI Assistance	USN-15	As a developer, I can integrate the Google Gemini SDK so the application can generate AI	5	High	Dillibabu J

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
			content.			
Sprint-3		USN-16	As a user, I can get a personalised AI workout recommendation from my age, goal and experience.	5	High	Dillibabu J
Sprint-3		USN-17	As a user, I can get AI fitness insights from my workout totals, average duration and calories burned.	5	Medium	Dillibabu J
Sprint-4	Testing & Documentation	USN-18	As a tester, I can use a Postman collection to test every API endpoint quickly.	3	Medium	Jagadeesh S
Sprint-4		USN-19	As a team member, I can refer to a README with setup steps and API reference.	2	Medium	Jagadeesh S
Sprint-4		USN-20	As a team, we carry out user acceptance testing and fix the bugs found.	5	High	Entire team

Project Tracker, Velocity & Burndown Chart: (4 Marks)

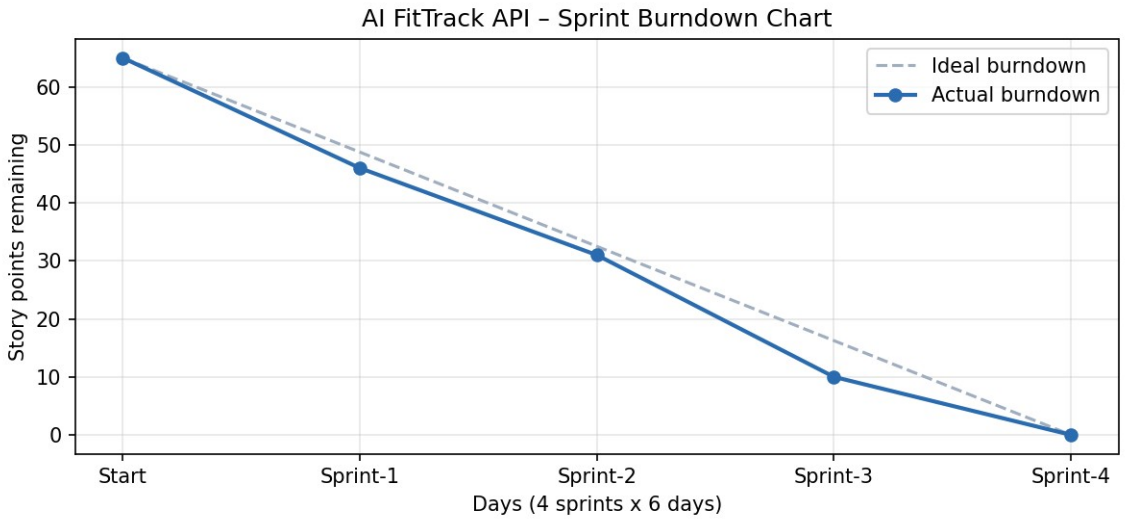
Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	19	6 Days	31 Aug 2026	05 Sep 2026	19	05 Sep 2026

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-2	15	6 Days	07 Sep 2026	12 Sep 2026	15	12 Sep 2026
Sprint-3	21	6 Days	14 Sep 2026	19 Sep 2026	21	19 Sep 2026
Sprint-4	10	6 Days	21 Sep 2026	26 Sep 2026	10	26 Sep 2026

Velocity:
Our sprint duration is 6 days and the team completed 65 story points across 4 sprints, so the team's average velocity is 16.25 story points per sprint.
The average velocity (AV) per iteration unit (story points per day) is calculated as follows:
AV = Velocity / Sprint Duration = 16.25 / 6 = 2.71 story points per day

Burndown Chart:

A burn down chart is a graphical representation of work left to do versus time. It is often used in agile software development methodologies such as Scrum. However, burn down charts can be applied to any project containing measurable progress over time.



https://www.visual-paradigm.com/scrum/scrum-burndown-chart/
https://www.atlassian.com/agile/tutorials/burndown-charts

Reference:

https://www.atlassian.com/agile/project-management
https://www.atlassian.com/agile/tutorials/how-to-do-scrum-with-jira-software
https://www.atlassian.com/agile/tutorials/epics
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